

Sketching Algorithms in the Age of Large Language Models

Blavatnik Fund Research Proposal

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1 Overview

Introduction. Large language models (LLMs) are revolutionizing society while computer science research on them continues to move fast. This is an exciting opportunity for many classical disciplines in the field: can the foundational knowledge and advances built and developed over many decades be brought to bear on and benefit this revolution?

Sketching algorithms in particular are a natural candidate. LLMs are infamous for the massive scale of computational resources they require, while massive work on scaling them up yet continues, since scale is believed to lead to qualitative boosts in LLM performance and to new emerging capabilities. As a consequence, computational efficiency in LLMs is a gateway not only to lower costs and increased availability, but also to core scientific progress.

Research objectives. The overarching theme of the proposed research is to leverage the rich foundational literature on sketching algorithms for significant progress in LLM efficiency and expansion of capabilities. As a computer science discipline, sketching algorithms are renowned as an attractive convergence point of rigorous mathematical foundations with plausible practicality and strong empirical performance. The premise of the current proposal is that their use in LLMs is still in its infancy and holds the potential for many innovative methods to discover and significant gains to be achieved.

Specifically, within the funding period, the research is planned to focus on two particular directions that have grown out of the PI's research over the last two years: *i* algorithmic foundations of positional encoding and *(ii)* population-level insights from user-LLM interactions. They are detailed in Section 2.

Methodology, logistics and personnel. The research is planned to be a combination of theoretical work with empirical implementation and experimentation. This is in line with the work done so far that has led to the preliminary results described in more detail in Section 2.

The requisite experiments are planned to run the department's GPU cluster, complemented by ad-hoc cloud compute as needed, a combination which has proven adequate so far for the desired scope. The majority of funding is intended for graduate student scholarships.

For details relating this proposal to a previous Blavatnik Fund grant, please refer to Section 3.

2 Detailed Description of the Proposed Research

2.1 Efficient Algorithmic Foundations for Positional Encoding

The Transformer architecture at the basis of LLMs has so far remained remarkably similar to its original form as presented in [20]. One aspect is an exception: positional encoding, which largely governs how the ordinal positions of tokens in the sequence inform their interaction. This has become increasingly important as the demand for ever longer context lengths in LLMs is growing.

The sinusoidal positional encoding scheme of the original Transformer in [20] is now mostly considered obsolete. Current LLMs (at least those that disclose their architecture) use a combination of newer techniques, including “chunked” attention, Attention with Linear Biases (ALiBi) [15], Local and Smooth Attention (LaS) [23], Rotary Position Encoding (RoPE) [16], and No Positional Encoding (NoPE) [10]. While much research has gone into speeding up the vanilla dot-product attention operation (e.g., [19, 18, 3, 11, 4, 1, 9], and more), new positional encoding schemes introduce their own algorithmic and computational bottlenecks into the mix. The proposed research is intended to address these emerging questions through the lens of rigorous sketching algorithms in order to speed up long-context LLMs.

Preliminary results. With Daniel Wolfson, an MSc student in the PI’s group, we have started research into the ALiBi method [15], as it seems particularly closely tied to sketching. We develop a fast sketching-based algorithm for Attention with ALiBi, which provably approximates the exact matrix while requiring only near-linear time to compute. Importantly, the algorithm appears highly practical, since it reduces ALiBi to a sequences of vanilla Attention computations on smaller matrices of varying sizes. Since the Attention operation is so fundamental to LLMs, it has been minutely optimized in its full form, and using it as a black-box in our algorithm allows our implementation to capitalize on existing optimized implementations like FlashAttention [6].

Preliminary experimental results show promise. We started with comparatively small LLMs that we are able to train from scratch on available compute. They achieve similar perplexity when with original ALiBi and with our approximate method, empirically confirming the validity of our approximation guarantee. However, context length of those LLMs is too short to exhibit efficiency gains – after all, our method is designed for long-context models. We are currently in the process scaling up our empirical evaluation to truly long-context models, where we hope and expect to see significant speed up with no accuracy loss.

Continued research. The above described work is still ongoing and is in early stages; completing the theoretical and experimental analysis is expected to occupy a significant portion of the funding period. The theory does not yet cover causal attention though we currently believe we have the tools to address that case. Thereafter, we plan to extend our method from textual sequences to images and, more intriguingly, Graph Attention, which seem compatible with the underlying theoretical approach. Our method also uncovers underlying connections between ALiBi and other commonly used positional encoding methods, and this connection remains to be explored.

2.2 Interpretable Population Level Insights from User-LLM interaction

LLMs have already permeated daily use to the extent that a large portion of web searches are made through LLMs instead of traditional search engines. This raises the question: can we draw population-level insights and analyze emerging trends from user-LLM interactions? Essentially, can we develop an analog of Google Trends for LLMs?

This relates to the fundamental question of interpreting LLM internal workings and reasoning. The way LLMs make sense of user prompts and decide on responses is currently considered mysterious and is a highly active topic of research. A major trend in this area has been Mechanistic Interpretability with Sparse Auto-Encoders (MI-SAEs). These are pre-trained models tailored to a particular LLM, and their role is to analyze intermediate layer LLM activations and extract from them interpretable features. These features yield interpretable insight into how the LLM “understands” the prompts it receives [5, 7, 14, 12, 2]. See illustration in Figure 1.

This alludes to an intriguing opportunity for sketching and streaming algorithms. Given a large collection of users interacting with an LLM, can we collect and analyze data from intermediate layer activations to draw population-level insights on the topics and trends of user-LLM interactions? Aggregation and analysis of this data would require sketching due to its dimensionality and scale. Since users interact with LLMs continuously over time, streaming algorithms may also be beneficial.

Preliminary results. With Barak Gorodisski, an MSc student in the PI’s group, we obtained initial results and they show promise. We took a 14-way text classification dataset (Wikipedia article summaries), ran each text through Gemma-2-2B (an open-parameter LLM by Google DeepMind), and extracted features with the LLM’s publicly available pre-trained MI-SAE. We designed a basic sketching scheme to score semantic concepts. Concepts aligned with true textual class (which is unknown to the algorithm) consistently and significantly outscore those aligned with other classes.

Continued research. This work is in early stages, and the current basic method needs to be developed and tested. Moreover, one of its main limitations is that it extracts MI-SAE features from each user-LLM interaction separately and then sketches the extracted features from all interactions. This imposes several limitations: (a) it makes the method more compute-heavy, (b) it ties it to a specific MI-SAE – if the MI-SAE is changed (say, improved or replaced), interactions already sketched cannot be re-analyzed, even if the LLM itself is unchanged. It would be preferable to

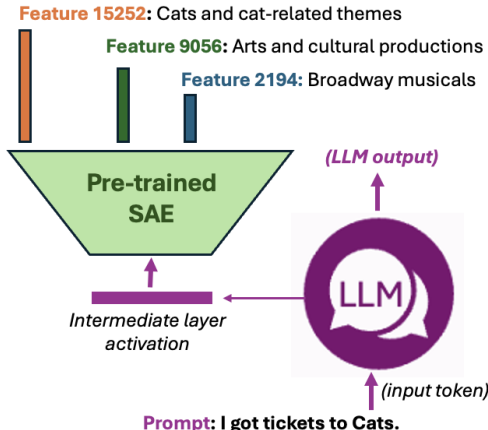


Figure 1: Illustration of SAEs in mechanistic interpretability of LLMs. Features shown are from Google DeepMind’s public pre-trained SAE for Gemma-2-2B, layer 24. The depicted example can be recreated with the Neuronpedia web interface ([link](#)).

have a sketching method that directly sketches intermediate layer activations, and applies the MI-SAE after sketching rather than before, and this appears to necessitate a different kind of sketching techniques. We are currently about to test an idea in this vein. Finally, since user-LLM interactions are prone to containing sensitive information, privacy needs to be taken into account when analyzing them at the population level, a matter recently addressed in [17, 13]. The sketching methods we consider seem highly compatible with the PI’s recent work on differentially private sketching and data analysis methods [22, 21] and we plan to explore them and incorporate rigorous user privacy.

3 Relation to Previous Blavatnik Fund Funded Research

The research proposed in this proposal largely grew out of work done in the PI’s group over the past two years, funded in large part by a Blavatnik Fund grant 2023-24, titled “*Breaking Algorithmic Barriers in Big Data: Scalable Machine Learning in High Dimensions*”. Some details on outcomes, use of funding, and continuation of research are provided below.

Efficient Algorithmic Foundations for Positional Encoding: Research under the Blavatnik 2023-34 grant focused on efficient computation of vanilla long-context Attention, not the positional encoding aspect.

- Publication: [9], published in ICLR 2025, and as a preliminary abstract in the First Workshop on Long-Context Foundation Models in ICML 2024 [8], where it won the Best Paper Award.

Relation to proposed research: While the above work did not address positional encoding at all, the proposed research puts it front and center, which turns out to lead to an Attention method which is both asymptotically faster and more empirically viable, through a very different technique.

- The preliminary results described in Section 2.1 were obtained by **Daniel Wolfson**, a full-time MSc student funded by the same Blavatnik Fund 2023-24 grant.

Interpretable Population Level Insights from User-LLM interaction: Research under the Blavatnik 2023-34 grant focused on privacy-preserving population-level analysis of data collected from distributed end-user devices.

- Publication: [21], published in ICLR 2025 with a Spotlight distinction.

Relation to proposed research: The proposed research draws on the resulting insights to develop a method tailored specifically to user-LLM interactions, capitalizing on the recent explosion of progress of pre-trained MI-SAEs.

- The preliminary results described in Section 2.2 were obtained by **Barak Gorodisski**, a full-time MSc student funded (in part) by the same Blavatnik Fund 2023-24 grant.

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